

MLFF CUEQ-DEFAULT1-HF2 TRAIN2 FP32 parity-ceiling hotfix

Gate: CUEQ-DEFAULT1-HF2

Release: mdstats 0.20.215a0

Architecture revision: 82

Problem

CUEQ-DEFAULT1-HF1 froze a TRAIN2-only FP32 parity floor of $rtol=1e-5$, $atol=2e-6$ from earlier selected-head MH-1 evidence. MPA-0/default workstation qualification later reported $E_{max}=2.384e-7$, $F_{max}=8.911e-6$, $S_{max}=1.660e-7$, and $D_{max}=2.883e-7$, while deterministic selection remained identical. The result is outside the historical TRAIN2 absolute floor only in the force channel.

Revised authority

The frozen TRAIN2 FP32 parity policy is:

```
rtol = 1e-5
atol = 1e-5
```

The generic ACCEL1 source/DATA6 FP32 policy remains $rtol=1e-5$, $atol=1e-6$. FP64 remains $rtol=1e-10$, $atol=1e-12$ for both authorities.

The TRAIN2 $1e-5$ value is a backend-equivalence ceiling for finite-precision differences caused by different FP32 reduction/accumulation order. It is not a scientific convergence, fitting, validation, or deployment-accuracy threshold and may not be reused as one.

Required invariants

1. All compared outputs must be finite.
2. Deterministic selection fingerprints must remain identical.
3. The ceiling is fixed; no observed failure may auto-expand it.
4. The generic source/DATA6 authority remains unchanged.
5. FP64 authority remains unchanged.
6. No silent fallback from requested CuEq TRAIN2 execution to e3nn is allowed.
7. The active parity-policy content digest is persisted and bound into FINAL-GPU1 preflight/handoff provenance.

Regression anchors

The MPA-0/default reported envelope with $F_{max}=8.911e-6$ must pass the TRAIN2 FP32 policy but fail the generic source/DATA6 FP32 policy when tested against a zero-reference absolute floor. A $1.0001e-5$ zero-reference difference must fail the TRAIN2 FP32 policy.